

SPAS 1st Biennial International Conference 2026

Data Analytics for Scientific Research & Innovation

Bridging Foundational Concepts to Hands-on Analysis

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Workshop Schedule (1 hr 45 mins)

Time Frame	Session Focus	Key Deliverables
00:00 - 00:35	Part I: Conceptual Foundations	Data types, hypothesis testing, test decision trees
00:35 - 01:25	Part II: Hands-on R Workflow	Data import, cleaning, summary statistics, visualization
01:25 - 01:45	Part III: Statistical Modeling & Interpretation	Hypothesis testing, ANOVA/Regression, Q&A session

Target Audience: Lecturers & Researchers across Statistics, Mathematics, Applied Chemistry, Microbiology, Biochemistry, Physics Electronics, Glass & Ceramics Tech, Hospitality Management, and Food Science.

Part I: Why Data Analytics in Pure & Applied Sciences?

Data analysis transforms raw laboratory measurements, survey responses, and experimental observations into **actionable scientific insight**.

Common Research Challenges

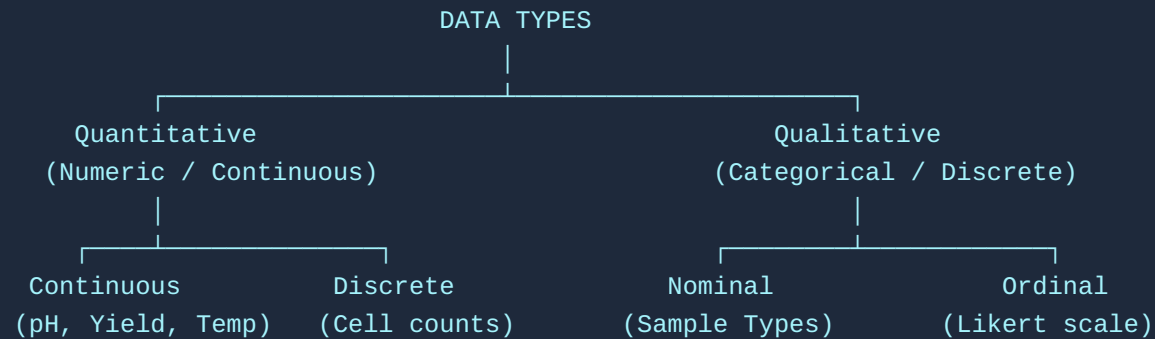
- Small sample sizes ($n < 30$) in lab trials
- Non-normal laboratory readings & outliers
- Distinguishing genuine signal from experimental noise
- Selecting the wrong statistical test for data types

Strategic Goal

To equip researchers across **Microbiology, Chemistry, Physics, Hospitality, and Food Science** with standard, reproducible statistical techniques using R and Quarto.

Understanding Data Types

Before choosing a statistical test, you **must** identify your measurement scale:



Rule of Thumb: Your data type directly dictates whether you apply **parametric tests** (e.g., t-test, ANOVA) or **non-parametric tests** (e.g., Mann-Whitney, Kruskal-Wallis).

Formulating Scientific Hypotheses

Every statistical test starts with two competing hypotheses:

1. Null Hypothesis (H_0)

- Statement of **no difference**, **no effect**, or **no association**.
- *Example (Food Tech): H_0* : Shelf-life duration does not differ across three storage packaging materials.

2. Alternative Hypothesis (H_1 or H_a)

- Statement of **an effect**, **a difference**, or **a relationship**.
- *Example (Food Tech): H_1* : At least one packaging material yields a significantly different shelf-life duration.

Statistical Test Selection Matrix

Research Question	Input (X)	Output (Y)	Parametric Test	Non-Parametric
Compare 2 independent groups	Categorical (2 levels)	Continuous (Normal)	Independent t-test	Mann-Whitney U test
Compare 2 paired measurements	Categorical (Paired)	Continuous (Normal)	Paired t-test	Wilcoxon signed-rank
Compare 3+ groups	Categorical (3+ levels)	Continuous (Normal)	One-Way ANOVA	Kruskal-Wallis test
Assess linear relationship	Continuous	Continuous	Pearson Correlation	Spearman Correlation
Predict continuous response	Continuous/Categorical	Continuous	Linear Regression	GLM / Non-linear

Interpreting p-values & Coefficient of Determination (R^2)

The p-value

- Measures the probability of observing your data under H_0 .
- Standard Alpha Threshold ($\alpha = 0.05$):
 - $p < 0.05$: Reject $H_0 \rightarrow$ Statistically significant effect observed.
 - $p \geq 0.05$: Fail to reject $H_0 \rightarrow$ Insufficient evidence of effect.

Coefficient of Determination (R^2)

- Quantifies the proportion of variance in the outcome (Y) explained by the predictor (X).
- *Example:* $R^2 = 0.84$ implies 84% of product viscosity change is explained by temperature variation.

Part II: Hands-on Workflow in R

Environment Setup & Package Loading

```
# Load required R packages for the workshop
library(tidyverse) # Data manipulation (dplyr) & visualization (ggplot2)
library(ggpubr)    # Publication-ready plots with stat p-values
library(rstatix)   # Pipe-friendly statistical analysis tools

# Set global clean theme for plots
theme_set(theme_minimal(base_size = 14))
```

Simulating Multidisciplinary Dataset

```
set.seed(2026)
research_data <- tibble(
  Sample_ID = 1:60,
  Department = rep(c("Food_Science", "Applied_Chem", "Microbiology"), each = 20),
  Treatment_Temp = rep(c(25, 45, 65, 85), times = 15),
  Enzyme_Activity = round(c(rnorm(20, 45, 5), rnorm(20, 62, 6), rnorm(20, 55, 4)), 2)
)
```


Data Exploration & Visualizations in R

Computing Summary Statistics

```
# Calculate Mean, SD, Median, and IQR by Department
summary_table <- research_data %>%
  group_by(Department) %>%
  get_summary_stats(Enzyme_Activity, type = "common")

print(summary_table)
```

Exploratory Boxplot

```
# Visualization across groups
ggplot(research_data, aes(x = Department, y = Enzyme_Activity, fill = Department)) +
  geom_boxplot(alpha = 0.7, outlier.colour = "red") +
  geom_jitter(width = 0.1, alpha = 0.5) +
  labs(title = "Enzyme Activity Distribution Across SPAS Departments", y = "Enzyme Activity (U/mL)")
```

Part III: Inferential Statistics & Regression

One-Way ANOVA & Post-Hoc Tests

```
# Compute ANOVA test
anova_res <- research_data %>% anova_test(Enzyme_Activity ~ Department)
print(anova_res)

# Pairwise Tukey HSD Post-Hoc Test
post_hoc <- research_data %>% tukey_hsd(Enzyme_Activity ~ Department)
print(post_hoc)
```

Linear Regression Modeling

```
# Fit linear model predicting Enzyme Activity from Temperature
fit_model <- lm(Enzyme_Activity ~ Treatment_Temp, data = research_data)
summary(fit_model)
```

Practical Takeaways & Best Practices

1. **Know Your Data Scale:** Categorical vs. Continuous measurements strictly dictate your analytical pipeline.
2. **Check Model Assumptions:** Always check normality, variance homogeneity, and linearity before concluding p-values.
3. **Emphasize Scientific Meaning:** Pair p-values with effect sizes (η^2 , R^2) and confidence intervals.
4. **Adopt Reproducible Workflows:** Use Quarto (`.qmd`) documents in RStudio / VS Code to combine code, analytical outputs, and publication manuscripts seamlessly.

Thank You! Questions & Live Hands-on Practice

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